CNNs, Vision Transformers and a CNN–Transformer Hybrid for Gastrointestinal Endoscopy Classification: Accuracy, Cost, Interpretability and Shortcut Learning on Kvasir v2

[Author Name]
Master's Program in Artificial Intelligence
[University]
[City, Country]
[email]

***Abstract*—We compare four deep-learning architectures for the eight-class classification of gastrointestinal endoscopy images (Kvasir v2, 8,000 images) under a single experimental protocol: a classic CNN (ResNet-50), an efficient CNN (EfficientNet-B0), a pure Vision Transformer (ViT-S/16) and a CNN–Transformer hybrid built for this study (ResNet-50 stages 1–3 followed by four Transformer layers), plus a soft-voting ensemble selected on the validation set. Besides accuracy and macro-F1 with bootstrap confidence intervals and McNemar tests, we report parameters, MACs, training time, GPU memory, latency and convergence, a one-factor-at-a-time hyperparameter study, Grad-CAM and attention rollout maps, and a shortcut-learning test based on an acquisition artifact. All models exceed 0.90 macro-F1. ViT-S/16 is the best single model (0.936, 93.6 % accuracy) and is significantly better than the other three, but it was also pretrained on ten times more data. Pretraining dominates every architectural choice: training from scratch costs 0.19 to 0.26 macro-F1, six to nine times the spread between the pretrained models, and explains why the hybrid, with 47 % of its weights randomly initialized, is the weakest model. EfficientNet-B0 reaches 97 % of the best score with 1/12 of the compute. Most remaining errors confuse esophagitis with the normal Z-line, and every model partly relies on a colonoscopy-only display box (0.014–0.034 macro-F1 lost when it is masked).**

***Index Terms*—medical image classification, gastrointestinal endoscopy, Kvasir, convolutional neural networks, vision transformers, transfer learning, interpretability, shortcut learning**

I. Introduction

Endoscopy is the reference examination of the gastrointestinal (GI) tract, but its yield depends on the attention and experience of the endoscopist: subtle lesions and early inflammation are missed, and the interpretation of findings varies between observers. Computer-aided detection and classification systems can act as a second reader, and deep learning has become the dominant approach since the release of public datasets such as Kvasir [1] and HyperKvasir [2].

For a decade, convolutional neural networks (CNNs) such as ResNet [3] and EfficientNet [4] were the default choice. Vision Transformers (ViT) [5] replace convolutions with global self-attention and lead natural-image benchmarks when enough pretraining data is available, while hybrids combine a convolutional stem with Transformer layers [5]–[7]. Whether these advances carry over to small medical datasets is not obvious: Transformers lack the locality bias of convolutions and depend heavily on pretraining [8], [9].

This work answers four questions on Kvasir v2 under a single, fully reproducible protocol: (i) which family works best on a small dataset (5,600 training images) and why; (ii) what each model costs and how fast it converges; (iii) where the models look and whether they exploit shortcuts; and (iv) how sensitive the conclusions are to the main hyperparameters. Our contributions are:

- A compute-matched comparison of ResNet-50, EfficientNet-B0 and ViT-S/16, and a CNN–Transformer hybrid built for this study, with significance tests and a complete cost profile (parameters, MACs, time, memory, latency, convergence).
- A one-factor-at-a-time sensitivity study (learning rate, batch size, resolution, augmentation, pretraining), including from-scratch training of a CNN and a Transformer.
- The identification and quantification of a shortcut in Kvasir v2: a display box present only in colonoscopies, on which every model partly relies.

II. Related Work

GI endoscopy datasets and benchmarks. Kvasir [1] introduced eight classes of anatomical landmarks, pathological findings and endoscopic procedures annotated by endoscopists; its second version, used here, has 1,000 images per class. The MediaEval Medico tasks [10] built on it and established transfer learning from ImageNet as the dominant approach. HyperKvasir [2] extended it to 23 classes and more than 100k unlabeled frames, and Kvasir-SEG [11] added polyp segmentation masks. Thambawita *et al.* [12] showed that models trained on one GI dataset generalize poorly to another, which motivates the analysis of dataset-specific cues in this work.

CNNs and Transformers in medical imaging. CNNs pretrained on ImageNet have been the standard in medical

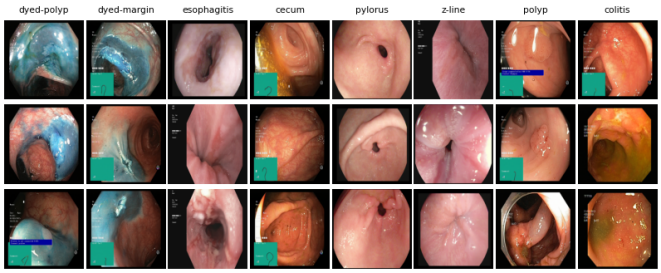


Fig. 1. Kvasir v2 examples (three per class). The teal box in the lower-left corner of several colon images is the ScopeGuide display (Section IV-F).

image analysis [13], although Raghu *et al.* [14] showed that the benefit of transfer learning shrinks for large medical datasets. ViT [5] needs large-scale pretraining; DeiT [15] and AugReg [8] reduced this need with stronger training recipes, and Swin [16] reintroduced locality through shifted windows. In medical imaging, Matsoukas *et al.* [9] found that ImageNet-pretrained ViTs match CNNs while CNNs win when both are trained from scratch, and Shamshad *et al.* [17] review the rapid adoption of Transformers. Hybrid designs, such as the ResNet–ViT hybrid of the original ViT paper [5], TransUNet [6] and CoAtNet [7], combine convolutional feature extraction with global attention.

Interpretability and shortcuts. Grad-CAM [18] and attention rollout [19] are the usual tools to inspect where CNNs and Transformers look. Shortcut learning [20], where a model exploits a spurious cue correlated with the label, has been documented in medical imaging, e.g. for COVID-19 radiographs [21].

III. Methodology

A. Dataset and split

Kvasir v2 contains 8,000 JPEG images (720×576 to 1920×1072 pixels) in eight balanced classes: dyed-lifted polyps, dyed resection margins, esophagitis, normal cecum, normal pylorus, normal Z-line, polyps and ulcerative colitis (Fig. 1). We use a stratified 70/15/15 split (5,600/1,200/1,200 images, seed 42). The validation set drives early stopping and the ensemble selection; the test set is used once. Because the classes are balanced, no re-weighting or re-sampling is needed.

B. Preprocessing and augmentation

Images are resized once to 256×256 and cached. Training uses a random 224×224 crop, horizontal and vertical flips, rotations of ±15° and brightness/contrast changes of ±10 %; validation and test use a center crop. Flips and rotations are valid because an endoscope has no canonical orientation. Hue and saturation are *not* altered because color is diagnostic: the indigo-carmine dye separates dyed from non-dyed polyps, and redness indicates inflammation. Each model is normalized with the statistics of its own pretraining (ImageNet mean/std for the CNNs, 0.5/0.5 for the AugReg ViT). Augmentation runs on the GPU with a single affine resampling per batch.

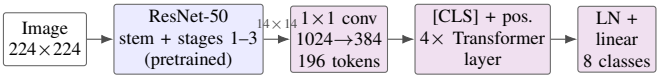


Fig. 2. The hybrid built in this work. Blue: pretrained weights (LR 10^{-4}); violet: new layers trained from scratch (LR 10^{-3}).

C. Models

ResNet-50 [3] (23.5 M parameters, 4.1 GMACs). ResNet-18/34 have less capacity, and ResNet-101/152 cost two to three times more compute for one to two ImageNet points, a gain unlikely to transfer to 5,600 images. ResNet-50 is the usual CNN baseline in the Kvasir literature. Weights: torchvision's improved recipe.

EfficientNet-B0 [4] (4.0 M, 0.38 GMACs). EfficientNet scales depth, width and input resolution jointly; B0 is the only variant designed for 224 px, which keeps the input identical for all models.

ViT-S/16 [5] (21.7 M, 4.6 GMACs), pretrained on ImageNet-21k with the AugReg recipe [8]. Its compute matches ResNet-50, so their comparison isolates convolution versus self-attention rather than budget. ViT-Ti is too small, ViT-B is four times more expensive, and Swin-T's local windows would blur the CNN–Transformer contrast.

Hybrid ResNet-50 + Transformer (16.1 M, 4.9 GMACs), built for this study (Fig. 2). The pretrained ResNet-50 stem and stages 1–3 produce a 14×14×1024 map that is projected to 384 channels, flattened into 196 tokens, prepended with a class token and added to learned position embeddings, and processed by four pre-norm Transformer layers (6 heads, MLP 1536, dropout 0.1). The design follows the ViT hybrid [5] and the TransUNet encoder [6]: the CNN extracts local texture, attention relates distant regions. Cutting after stage 3 keeps 196 tokens (stage 4 would leave 49) and makes the hybrid a controlled variant of ResNet-50 in which stage 4 and global pooling are replaced by attention. Its 7.6 M new parameters (47 %) are randomly initialized.

D. Training protocol

All models share the protocol of Table I. The same seed fixes initialization, batch order and augmentation. Training runs on one NVIDIA L4 GPU with bfloat16 mixed precision.

E. Evaluation

The primary metric is test macro-F1, the unweighted mean of the per-class F1 scores, which prevents a weak class from hiding behind strong ones; we also report accuracy, per-class F1 and confusion matrices. 95 % confidence intervals are obtained with 1,000 paired bootstrap resamples of the test set [26], and pairs of models are compared with the exact McNemar test [27], [28]. Cost is measured as parameters, multiply-accumulate operations (MACs), training time, peak GPU memory, throughput (batch 64) and single-image latency. Convergence is measured as the best epoch and the number of epochs to reach 95 % of the best validation macro-F1. Success criteria fixed before training: every model ≥0.90 macro-F1;

TABLE I
SHARED TRAINING PROTOCOL AND RATIONALE

Setting	Value	Rationale
Input	224 px	Native resolution of all pretrained weights
Optimizer	AdamW	Decoupled weight decay [22]; standard for CNN and ViT fine-tuning
Learning rate	10^{-4}	Fine-tuning; hybrid's new layers 10^{-3}
Schedule	1-epoch warm-up + cosine	Protects pretrained weights from the random head's early gradients [23], [24]
Batch size	32	175 updates per epoch; small-batch noise regularizes
Weight decay	0.05	Not applied to biases, norms, position embeddings
Label smoothing	0.1	Less over-confidence, better-calibrated probabilities [25]
Epochs	$\leq$20, patience 4	Early stopping on validation macro-F1
Grad. clipping	norm 1.0	Guards against gradient spikes

differences claimed only if $p < 0.05$; the best model loses $\leq$0.02 macro-F1 when the shortcut is removed.

Ensemble. All 11 soft-voting combinations of two or more models are scored on the validation set; the best (ties broken by size) is evaluated once on the test set.

Sensitivity. Starting from the EfficientNet-B0 configuration, each run changes one factor: learning rate (10^{-5}, 10^{-3}), batch size (128, same LR), resolution (64 px), no augmentation, and random initialization. One extra run trains ViT-S/16 from scratch.

Interpretability and shortcut test. Grad-CAM [18] is computed on the last convolutional map of each CNN (and of the hybrid's CNN part) and attention rollout [19] on the Transformer layers. The ScopeGuide box is located from the data (teal pixels: $G > 100$, $G - R > 60$, $G > B$), masked in the test images, and the models are re-evaluated without retraining. We also compute the share of each explanation map that falls inside the box, divided by the box's share of the image area (1 = uniform).

IV. RESULTS

A. Overall performance

Table II summarizes the test results. All models meet the 0.90 criterion. ViT-S/16 is the best single model (macro-F1 0.936) and is significantly better than ResNet-50 ($p = 0.041$), EfficientNet-B0 ($p = 0.002$) and the hybrid ($p < 0.001$). The three CNN-based models are not significantly different from each other ($p = 0.07$–0.54). The ensemble selected on validation, ViT-S/16 + Hybrid, has the highest point estimate (0.938), but its gain over ViT-S/16 is not significant ($p = 0.63$; 10 versus 7 discordant images). The full run took 44.7 minutes.

B. Cost and convergence

Fig. 3 relates performance and cost. EfficientNet-B0 reaches 97 % of the best macro-F1 with about one fifth of ViT-S's parameters and one twelfth of its MACs, and has the best

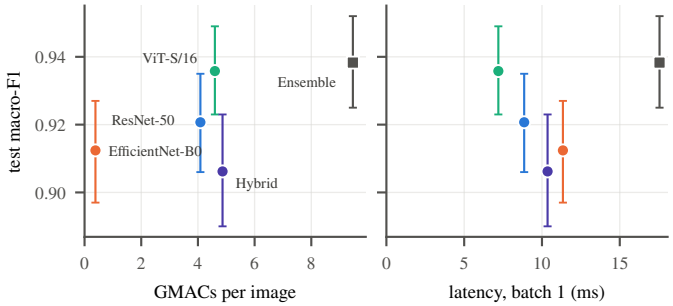


Fig. 3. Test macro-F1 (95 % CI) versus compute and single-image latency. Fewer MACs do not imply lower latency.

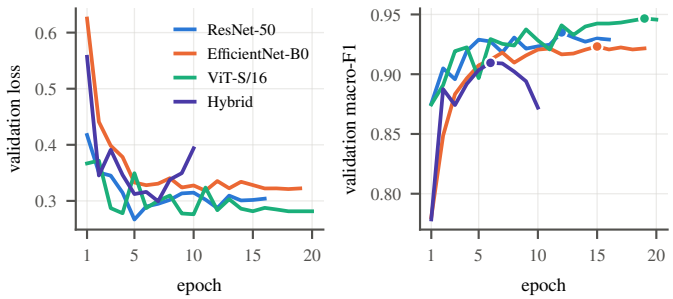


Fig. 4. Validation loss and macro-F1 per epoch; dots mark the epoch whose weights were kept.

batched throughput. Yet, with a single image, it is the slowest model (11.3 ms versus 7.2 ms for ViT-S/16): its many small depthwise layers are bound by memory access and kernel launches rather than arithmetic, whereas ViT-S maps to a few large matrix products. All models are far below the 33 ms frame budget of a 30 fps video.

All models reach 95 % of their best validation macro-F1 within two to three epochs (Fig. 4), a consequence of pretraining. ViT-S/16 kept improving until epoch 19, using almost the whole cosine schedule. The hybrid peaked early (epoch 6): its validation loss rose from 0.30 to 0.39 between epochs 7 and 10 while its training loss stayed near 0.63 (about 0.48 for the other models), a sign of unstable optimization of its new layers.

C. Per-class performance and errors

Pylorus, cecum and ulcerative colitis are nearly solved (Table III). Esophagitis and normal Z-line are the only classes below 0.90 F1 for every model, and their mutual confusion is the most frequent error (121 esophagitis images predicted as Z-line and 83 Z-line images predicted as esophagitis, summed over the four models). ViT-S/16 is the best on this pair (esophagitis recall 0.88), EfficientNet-B0 the worst (0.77). The hybrid has the best Z-line recall (0.92) at the cost of the worst esophagitis recall (0.73), and its errors overlap the least with those of ViT-S/16 (Jaccard index 0.33, the lowest of all pairs), which explains its selection for the ensemble. The second confusion, dyed resection margins versus dyed-lifted polyps, depends on whether the polyp has already been removed.

TABLE II

TEST RESULTS (1,200 IMAGES). MACRO-F1 AND ACCURACY WITH 95 % BOOTSTRAP CIS; MCNEMAR p AGAINST VIT-S/16. COST OF THE ENSEMBLE IS THE SUM OF ITS MEMBERS. LATENCY: ONE IMAGE; THROUGHPUT: BATCH OF 64 (NVIDIA L4, BF16).

Model	Macro-F1 [95 % CI]	Acc. (%)	p vs ViT	Params (M)	GMACs	Train (s)	Memory (MB)	Latency (ms)	Thr. (ms/img)	Best / run ep.
Ensemble ViT-S + Hybrid	**0.938** [0.925, 0.952]	**93.83**	0.63	37.8	9.47	353	–	17.56	1.26	–
ViT-S/16	0.936 [0.923, 0.949]	93.58	–	21.7	4.60	190	1545	**7.20**	0.45	19 / 20
ResNet-50	0.921 [0.906, 0.935]	92.08	0.041	23.5	4.09	246	1909	8.86	0.74	12 / 16
EfficientNet-B0	0.912 [0.897, 0.927]	91.25	0.002	**4.0**	**0.38**	210	**1492**	11.35	**0.39**	15 / 19
Hybrid R50 + Transformer	0.906 [0.890, 0.923]	90.67	<0.001	16.1	4.87	**162**	2089	10.36	0.81	6 / 10

TABLE III

TEST F1 PER CLASS

Model	D-pol	D-mar	Esoph	Cecum	Pylor	Z-line	Polyp	Colit
Ensemble	0.94	0.94	**0.87**	0.98	0.99	**0.86**	0.95	0.97
ViT-S/16	0.94	0.94	0.86	0.98	0.99	0.85	0.96	0.97
ResNet-50	0.93	0.94	0.84	0.97	0.98	0.84	0.92	0.95
EffNet-B0	0.93	0.94	0.79	0.97	0.99	0.80	0.93	0.94
Hybrid	0.91	0.92	0.81	0.94	0.99	0.84	0.91	0.94

TABLE IV

SENSITIVITY STUDY: CHANGE IN TEST MACRO-F1 RELATIVE TO THE PRETRAINED BASELINE (EFFICIENTNET-B0: 0.912; VIT-S/16: 0.936), WITH PAIRED BOOTSTRAP 95 % CIS

Run	Change	Macro-F1	Δ [95 % CI]
B0	LR $10^{-4} \to 10^{-5}$	0.806	−0.107 [−0.129, −0.086]
B0	LR $10^{-4} \to 10^{-3}$	0.927	+0.015 [+0.002, +0.030]
B0	batch 32→128	0.898	−0.014 [−0.029, −0.002]
B0	224→64 px	0.859	−0.053 [−0.074, −0.034]
B0	no augmentation	0.913	+0.001 [−0.013, +0.013]
B0	from scratch	0.654	−0.259 [−0.287, −0.233]
ViT-S	from scratch	0.749	−0.187 [−0.211, −0.162]

D. Hyperparameter sensitivity

Table IV and Fig. 5 show the one-factor-at-a-time study. Pretraining is by far the largest effect: EfficientNet-B0 and ViT-S/16 lose 0.259 and 0.187 macro-F1 when trained from scratch. A learning rate of 10^{-5} leaves the model under-fitted after 20 epochs, and 64 px inputs cost 0.053, confirming that fine detail matters. Two predictions failed: 10^{-3} *improved* EfficientNet-B0 (+0.015), so the shared learning rate was conservative for it, and removing augmentation had no measurable effect within 20 epochs.

E. Interpretability

Fig. 6 shows the explanation maps. ResNet-50 produces the sharpest Grad-CAM maps, centered on the dye, the pyloric opening or the inflamed junction; EfficientNet-B0's maps are broader. The ViT-S/16 rollout is diffuse and spreads over the image, and in the dyed-margin example it highlights the ScopeGuide box. The hybrid's rollout is the most focused, often on a few lesion patches. On the esophagitis→Z-line failures (Fig. 7) every model looks at the gastro-esophageal junction: the errors come from the subtle visual difference, not from looking at the wrong place.

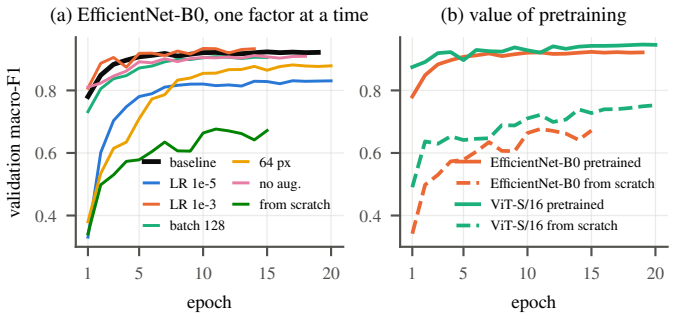


Fig. 5. Validation macro-F1 of the sensitivity runs. Dashed: training from scratch.

TABLE V

SHORTCUT TEST: MACRO-F1 CHANGE WHEN THE BOX IS MASKED, SHARE OF CHANGED PREDICTIONS ON THE 452 TEST IMAGES WITH THE BOX, AND EXPLANATION-MAP MASS IN THE BOX RELATIVE TO A UNIFORM MAP

Model	Δ macro-F1 [95 % CI]	Changed (%)	Map ratio
ResNet-50	**−0.014** [−0.024, −0.004]	**6.6**	**1.12**
EfficientNet-B0	−0.028 [−0.042, −0.013]	10.6	2.06
ViT-S/16	−0.025 [−0.038, −0.013]	8.0	1.67
Hybrid	−0.034 [−0.047, −0.021]	12.6	0.79 / 1.89
Ensemble	−0.026 [−0.038, −0.014]	8.4	–

F. Shortcut learning

The ScopeGuide display (a 3-D position indicator of the colonoscope) appears in 38 % of the images: in about 98 % of cecum images, in a third to two thirds of the polyp, dyed and colitis images, and in none of the upper-GI classes (Fig. 8). Its presence thus reveals "colonoscopy". Masking it in the test images lowers the macro-F1 of every model, with CIs that exclude zero (Table V). ResNet-50 is the most robust; the hybrid, whose Transformer part attends to the box (rollout ratio 1.89) while its CNN part does not (0.79), is the most affected. ViT-S/16 loses 0.025, slightly missing the success criterion of 0.02.

V. DISCUSSION

Why ViT-S/16 won. It combines global attention, tensor-core-friendly computation and, crucially, pretraining on ImageNet-21k (14 M images) against ImageNet-1k (1.3 M) for the CNNs. Our design matched compute, not pretraining data, so the result should be read as "a well-pretrained Transformer" rather than "attention beats convolution", consistent with [9]. It is also the best on the fine-grained esophagitis/Z-line pair

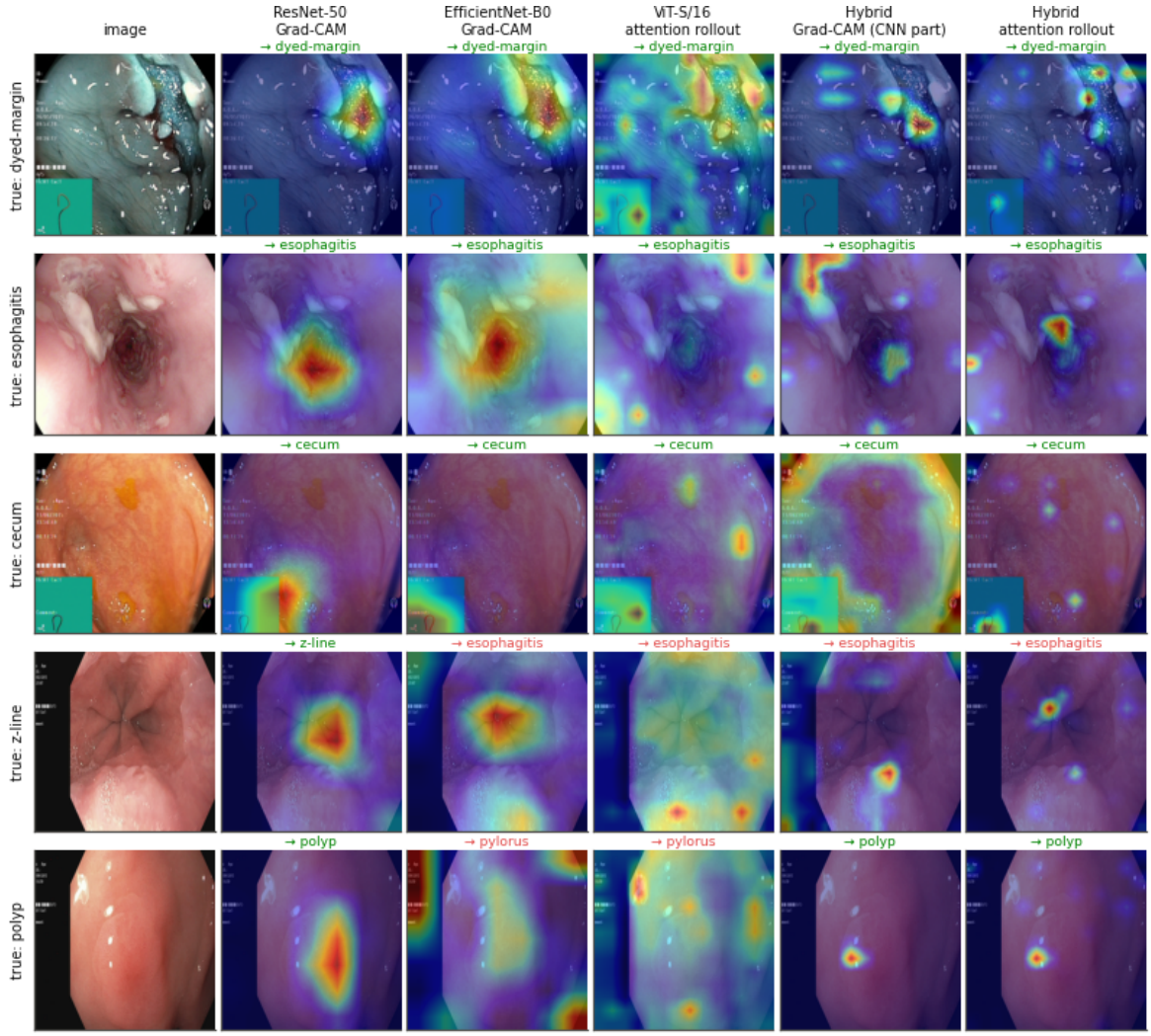


Fig. 6. Grad-CAM (CNNs, hybrid's CNN part) and attention rollout (ViT-S/16, hybrid's Transformer layers) on test images of five classes; the label above each map is the prediction (green: correct, red: wrong).

and the fastest per image, but its diffuse rollout maps are the hardest to explain to clinicians and it uses the box shortcut.

Why the hybrid failed as a single model. It discards the pretrained stage 4 of ResNet-50 and replaces it with 7.6 M random weights; on 5,600 images the lost pretrained features outweigh the added global modelling, the same lesson as the from-scratch runs. Its new layers, trained at a ten times larger learning rate, made validation loss rise after epoch 7, and its attention learned the box shortcut. Yet its errors are the most complementary to ViT-S/16's, which makes it a useful ensemble member.

Why ResNet-50 and EfficientNet-B0 behave differently. Both are statistically tied, but they suit different purposes. ResNet-50 has the most localized explanations and the smallest shortcut dependence. EfficientNet-B0 has a very low arithmetic cost but less capacity, which hurts on the hardest pair, and its latency advantage only appears in batched inference. Its improvement at 10^{-3} shows that one shared learning rate slightly under-rates it.

Which model for which use. For the best single-model accuracy and for real-time GPU inference: ViT-S/16. For low-power or edge hardware: EfficientNet-B0, which gives up 0.024 macro-F1 for twelve times fewer MACs. For explanations shown to clinicians and robustness to the artifact: ResNet-50. The ensemble is only justified for offline second reading, where its small gain is worth twice the compute.

Limitations. A single training seed (the CIs reflect test-set variability, not training variability); unequal pretraining data between ViT and CNNs; one shared learning rate; a single dataset from one centre without external validation; early stopping truncating the cosine schedule; a sensitivity study run one factor at a time on one model, with the fine-tuning recipe for the from-scratch runs; and masking introduces a small distribution shift, so the shortcut effect is an upper bound.

VI. Conclusion and Future Work

On Kvasir v2, all four architectures exceed 0.90 macro-F1. A pretrained ViT-S/16 is the most accurate and fastest

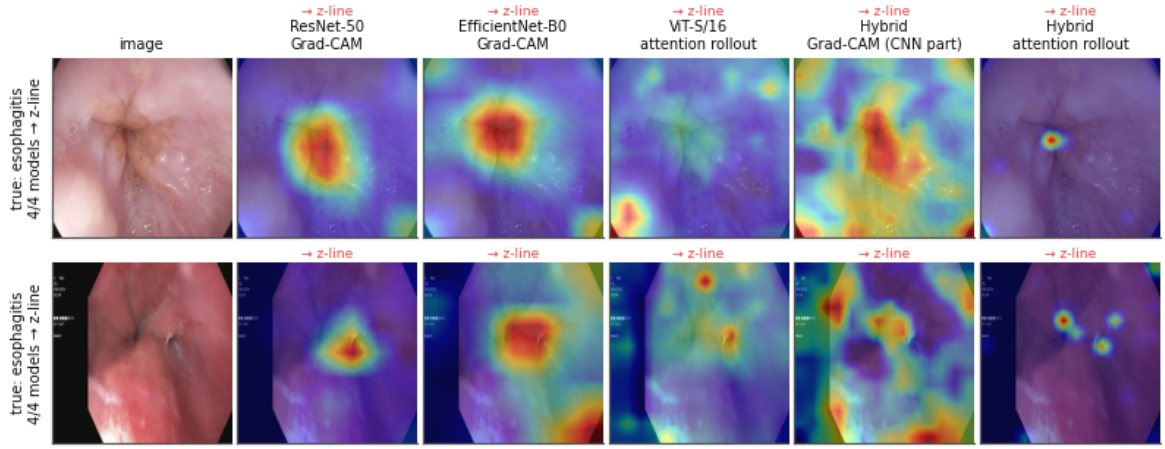


Fig. 7. Esophagitis images misclassified as normal Z-line by all four models: the maps are centered on the junction.

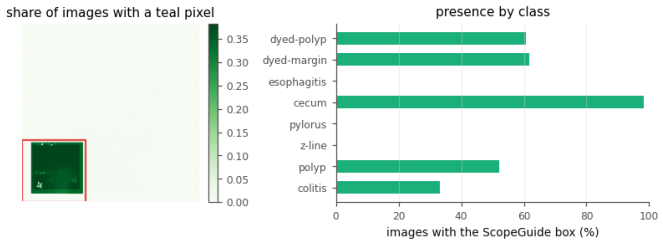


Fig. 8. Left: share of images with a teal pixel at each position (box region in red). Right: share of images of each class containing the ScopeGuide box.

single model, but the dominant factor is pretraining rather than architecture: training from scratch costs 0.19–0.26 macro-F1, six to nine times the spread between pretrained models, and a hybrid with half of its weights randomly initialized underperforms its own CNN backbone. Efficiency depends on the deployment scenario, the remaining errors are fine-grained (esophagitis versus Z-line), and every model partly exploits a colonoscopy-only display box. Future work includes comparing CNNs pretrained on ImageNet-21k (e.g. BiT [29]) to separate architecture from pretraining, self-supervised pretraining on the unlabeled HyperKvasir frames, initializing the hybrid's Transformer layers from pretrained weights, removing the ScopeGuide box during training with validation on other centres, higher resolution and grade-level labels for esophagitis, and distilling the ensemble into a real-time model.

Reproducibility

Code, the executed notebook and instructions are available at https://github.com/Yeldebao/AI-MASTERS/tree/main/deep-learning-kvasir. The notebook downloads the dataset, trains every model and regenerates all tables and figures in about 45 minutes on one NVIDIA L4 GPU (Google Colab).

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